AEGISPAY

LangGraph Architecture — GROW (Merchant Revenue Agent)

How an AI looks at a merchant's sales and safely invents offers that make money — **without giving itself an unlimited budget**.

1. What this graph does (in plain English)

A merchant sells shoes. The GROW agent reads the sales history, notices “customers who buy running shoes often buy running socks”, and proposes a small cross-sell offer. But it cannot switch the offer on by itself — a merchant approves it, and it runs only inside a fixed budget.

The most important idea

The AI is the idea person. It proposes. A merchant decides. And iron rules (max discount, max budget, minimum margin) are always enforced — no matter how confident the AI sounds.

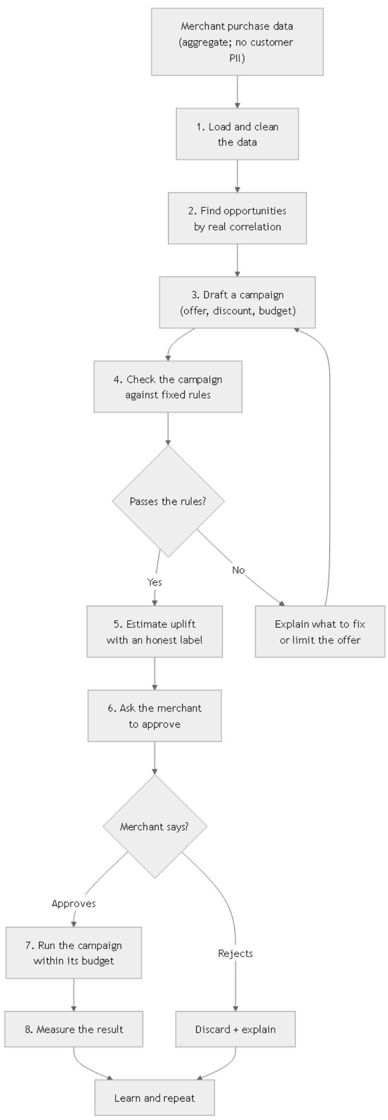

Figure 1 — The GROW agent graph: from merchant data, through rules and approval, to a measured campaign

2. What happens at each step

#	Node	What it does	Minder (the safety rail)
1	Load data	Pulls grouped sales history. No personal customer data needed.	Uses aggregate numbers only (affinity of product pairs, not who bought).
2	Find opportunities	Finds real correlations (shoe + socks, bottle + shirt).	Uses math (affinity, confidence), not a guess.
3	Draft campaign	Writes the offer: a discount, a budget, a target audience.	It only proposes. It cannot switch it on.
4	Check rules	Tests the offer against fixed limits.	Max discount, max budget, minimum margin, duration, email frequency.
5	Estimate impact	Predicts a possible lift, clearly labelled as an estimate.	It never claims a fake number; the estimate is a range, not a promise.
6	Approve	The merchant sees the why and says yes or no.	A merchant (or a policy) decides; the AI cannot approve itself.
7	Run campaign	Turns the offer on inside its budget.	The moment the budget is spent, it stops by itself.
8	Measure	Counts what actually happened.	If the offer did not help, it learns and does not repeat blindly.

3. The rules that stop bad AI ideas

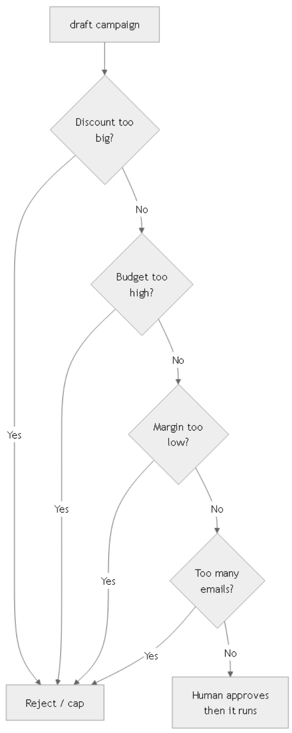

Figure 2 — Every proposed offer is filtered by fixed limits before a human ever sees it

Risk	The rule that stops it
Discount too big	A hard maximum discount percent.
Budget too high	A hard maximum campaign budget; it auto-stops when spent.
Eating into profit	A minimum margin floor - the offer must still make money.
Spamming customers	A limit on how many times one customer is contacted.
Unfair targeting	Only value/behaviour segments are allowed; never protected personal traits.

4. Built to run in production

&#8226**Propose, never force.** The graph always ends at a proposal that needs a merchant's yes.

&#8226**Hard caps, always.** Discount, budget, margin, duration and frequency are deterministic rules the AI cannot change.

&#8226**Stops when it should.** If the budget is used up, the campaign pauses by itself. No overspend.

&#8226**Honest numbers.** Revenue predictions are labelled as estimates with a range, never a fake confidence.

&#8226**Audited end-to-end.** Every proposal, approval and action is written to the audit ledger.

&#8226**Safe against abuse.** The red-team tries to make the AI invent a giant discount or a huge campaign, and the tests make sure it is blocked.

&#8226**Improves, doesn't burn money.** It measures results and only repeats what actually worked.

5. What you get

A merchant earns more without risking the business: the AI spots and proposes real opportunities, and the merchant stays in control of every rupee. The AI grows the store — it does not gamble with it.